



Jungseob Lee

NLP Researcher · Graduate Student, Korea University

📅 1995.10.11

✉ omanma1928@korea.ac.kr

🌐 js-lee-ai.github.io

🔄 js-lee-AI

🌐 LinkedIn

🎓 Google Scholar

Summary

Integrated M.S./Ph.D. researcher at Korea University NLP Lab (Advisor: Prof. Heuseok Lim), working on large language models and natural language processing — agentic LLMs, efficient reasoning, machine translation, information retrieval, and model merging. First-author work at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**; four papers accepted at **EMNLP 2026** (2 Main, 2 Findings). Served as post-training Project Manager (Korea University representative) for **VAETKI**, Korea's sovereign foundation model.

Research Interests: Large Language Models, Agentic LLM, Efficient Reasoning, Machine Translation, Information Retrieval, Model Merging

Education

Integrated M.S./Ph.D. in Computer Science

Korea University, Seoul, Korea

2021 – Present

Expected: Feb. 2027

- NLP Lab, Advisor: Prof. Heuseok Lim — large language models, agentic LLM, efficient reasoning, machine translation, information retrieval, and model merging

Research Experience

VAETKI — Korean Sovereign AI Foundation Model

Post-training Project Manager — Korea University Representative

Jun 2025 – Jan 2026

NC AI (NCSOFT)-led consortium

- Led the post-training pipeline (SFT, RL, dataset curation) as PM for Korea University; selected by the Ministry of Science and ICT for Korea's national Sovereign AI Foundation Model program (NC AI-led 14-institution consortium; open-sourced Dec 2025).
- State-of-the-art on Korean language/culture (CLiCk), reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [\[Model\]](#) [\[Technical Report\]](#)

Cross-lingual Language Transfer for Large Language Models

Lead Researcher (first author)

2024 – 2025

Published at ACL 2025 (Oral)

- Cross-lingual optimization that transfers English-centric LLMs to new languages while preserving source-language ability; **Oral** at ACL 2025.
- [\[PDF\]](#)

Length-aware Byte Pair Encoding for Korean Machine Translation

Lead Researcher (first author)

2024

Published at Findings of ACL 2024

- Length-aware BPE tokenization that mitigates over-segmentation in Korean machine translation and improves translation quality; **Findings of ACL 2024**.
- [\[PDF\]](#)

Awards

Best Paper Award (Top 1)

2025

Annual Conference on Human and Cognitive Language Technology (HCLT 2025)

For “*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*”; [J. Lee](#) (first author), M. Kim, J. Yoon, *et al.* — Korea University, Soongsil University, Human-Inspired AI Research.

Publications

Underlined = me; * equal contribution. Full list (31 papers) on [Google Scholar](#).

2026

- “*The Geometry of Hallucination Detection: Why Simple Probes Are Sufficient*” — [J. Lee](#), J. Seo, H. Lim **EMNLP 2026** (Main)
- “*Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization*” — S. Lee, M. Kim, [J. Lee](#), H. Lim **EMNLP 2026** (Main)
- “*DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models*” — [J. Lee](#), S. Hong, S. Lee, *et al.* **Findings of EMNLP 2026** [\[arXiv\]](#)
- “*ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models*” — M. Kang*, [J. Lee](#)*, J. Seo, H. Lim **Findings of EMNLP 2026**
- “*Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment*” — S. Hong, Y. Jang, [J. Lee](#), H. Moon, H. Lim **ICLR 2026** (Poster) [\[PDF\]](#)
- “*Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation*” — C. Ok, E.K. Lee, H. Ju, [J. Lee](#), D. Oh **Knowledge-Based Systems** [\[DOI\]](#)
- “*Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards*” — [J. Lee](#), S. Lee, S. Hong, M. Kim, C. Park, H. Lim Preprint [\[arXiv\]](#)
- “*To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG*” — [J. Lee](#), C. Park, H. Lim Preprint [\[arXiv\]](#)
- “*Skin-Deep: A Geometric Diagnostic for Alignment Fragility in LLM Representations*” — D.J. Lee*, [J. Lee](#)*, S. Lee, *et al.* Preprint [\[arXiv\]](#)

2025

- “*Cross-lingual Optimization for Language Transfer in Large Language Models*” — [J. Lee](#), S. Hong, H. Moon, H.S. Lim **ACL 2025** (Oral) [\[PDF\]](#)
- “*An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training*” — S. Son, C. Park, [J. Lee](#), *et al.* **Expert Systems with Applications** [\[DOI\]](#)

2024

- “*Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation*” — [J. Lee](#), H. Moon, S. Lee, C. Park, S. Eo, *et al.* **Findings of ACL 2024** [\[PDF\]](#)
- “*Analysis of Model-, Data-, and User-Centric Approaches for Chat Applications: A Case Study of BlenderBot 2.0*” — C. Park, [J. Lee](#), S. Son, *et al.* **Applied Sciences** [\[DOI\]](#)

2023

- “*A Survey on Evaluation Metrics for Machine Translation*” — S. Lee, [J. Lee](#), H. Moon, *et al.* **Mathematics** (142 citations) [\[PDF\]](#)
- “*PEEP-Talk: A Situational Dialogue-based Chatbot for English Education*” — S. Lee, Y. Jang, C. Park, [J. Lee](#), *et al.* **ACL 2023** (Demo) [\[PDF\]](#)

2022

- “*QUAK: A Synthetic Quality Estimation Dataset for Korean-English Neural Machine Translation*” — S. Eo, C. Park, H. Moon, J. Seo, G. Kim, [J. Lee](#), H.S. Lim **COLING 2022** [\[PDF\]](#)
- “*K-NCT: Korean Neural Grammatical Error Correction Gold-standard Test Set*” — S. Koo, C. Park, J. Seo, S. Lee, H. Moon, [J. Lee](#), H. Lim **IEEE Access** [\[PDF\]](#)

Last updated: August 24, 2026